Experiment-11

Write A Python Program For Map Coloring To Implement CSP

Aim

To write and execute a Python program to implement Map Coloring using a Constraint Satisfaction Problem (CSP), ensuring that no two neighboring regions have the same color.

Algorithm

1. Start the program.
2. Define the map regions and their neighboring regions.
3. Initialize the available colors as Red, Green, and Blue.
4. Create an empty dictionary to store color assignments.
5. Select an uncolored region.
6. Choose a color from the available colors.
7. Check whether the selected color conflicts with any neighboring region.
8. If there is no conflict, assign the color to the region.
9. Move to the next uncolored region.
10. If a conflict occurs, try another color.
11. Use backtracking if no valid color is available.
12. Display the color assigned to each region and stop the program.

Code

ex-11.py - D:/ex-11.py (3.10.0)

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```
# Map Coloring using CSP

# Define neighboring regions
graph = {
    'A': ['B', 'C'],
    'B': ['A', 'C', 'D'],
    'C': ['A', 'B', 'D'],
    'D': ['B', 'C']
}

# Available colors
colors = ['Red', 'Green', 'Blue']

# Store colors assigned to regions
color_map = {}


def is_safe(region, color):
    for neighbor in graph[region]:
        if color_map.get(neighbor) == color:
            return False
    return True


def solve_csp(regions):
    if not regions:
        return True

    region = regions[0]

    for color in colors:
        if is_safe(region, color):
            color_map[region] = color

            if solve_csp(regions[1:]):
                return True

            del color_map[region]

    return False


# Solve the map coloring problem
if solve_csp(list(graph.keys())):
    print("MAP COLORING SOLUTION")
    print("---------------------")

    for region, color in color_map.items():
        print(region, ":", color)
else:
    print("No solution exists.")
```

Output

IDLE Shell 3.10.0

File Edit Shell Debug Options Window Help

```
Python 3.10.0 (tags/v3.10.0:b494f59, Oct  4 2021, 19:00:18) [MSC v.1929 64 bit (
AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
============================== RESTART: D:/ex-11.py ==============================
MAP COLORING SOLUTION
---------------------
A : Red
B : Green
C : Blue
D : Red
>>>
```

Result

The Map Coloring problem was successfully implemented using CSP and backtracking in Python 3.10. All four regions were colored so that no two neighboring regions have the same color.